Technical Synopsis

For Independent Scientific Review

1. Scientific question

Can a single effective gravitational-saturation sector generate a nonsingular early history and a late vacuum-like residual while remaining compatible with major background and linear cosmological observables?

2. Degrees of freedom

The homogeneous state vector is (a, H, C, Pi_C, phi, Pi_phi, rho_m, rho_r). C is the organizational field and phi the phase field. Both are canonical in the displayed matter-sector action.

3. Potential

V(C,phi) = V_infinity + (m_C^2/2)(C-C_s)^2 + (lambda/4)(C-C_s)^4 + A_0(1-cos phi). The joint minimum is (C_s,0), where the residual equals V_infinity and the local scalar curvatures are positive for the locked parameters.

4. High-density gravitational response

The effective background constraint is H^2 = (rho/3)[1-(rho/rho_s)^q], with q=1 in the locked implementation. It recovers H^2 approximately rho/3 for rho much less than rho_s and gives H=0 at rho=rho_s.

5. Transfer and conservation

During contraction, a one-sided rate Gamma_tr cancels anti-friction and adds positive damping. Scalar kinetic loss appears as a radiation source Q = Gamma_tr(Pi_C^2+Pi_phi^2). Adding scalar, radiation, and matter continuity equations yields total conservation.

6. Bounce treatment

The evolution through the bounce is performed in cosmic time or a nonsingular affine parameter. Variables divided by H, such as derivatives with respect to N=ln a, are not used at H=0. The Raychaudhuri-type equation is interpreted through the continuous limiting relation implied by the differentiated constraint and conservation law.

7. Expanding branch

After the bounce the transfer is set to zero, the canonical fields damp toward the stable minimum, and the residual approaches V_infinity with w_C near -1. The resulting H(z) is locked and normalized to E(0)=1 with H0=67.4 km/s/Mpc and Omega_m0=0.314114.

8. Perturbations

Canonical synchronous-gauge perturbations delta C and delta phi are evolved with metric-source terms and contribute through delta rho, delta p, and momentum density, with zero intrinsic scalar anisotropic stress. They are not represented as an imposed rescaling G to mu G.

9. Observational program

The same locked history is used for primary CMB spectra, the acoustic scale, drag horizon, BAO distances, linear growth, S8, and Planck 2018 CMB lensing. The manuscript reports a close acoustic-scale match, moderate S8, and a nine-bin lensing statistic better than a matched control.

10. Main independent checks required

Independent reviewers should reproduce the background trajectory, verify the constraint residual through the bounce, inspect the exact CAMB source modifications, rerun baseline and control chains, repeat likelihood calculations with nuisance parameters, and test sensitivity beyond one-at-a-time scans.

11. Falsification conditions

The framework would be undermined by failure to conserve energy numerically, inability to evolve regularly through H=0, unstable scalar perturbations, irreproducible locked observables, unacceptable primary-CMB or BAO likelihoods, or required retuning that defeats the frozen-parameter claim.

12. Present status

The manuscript is suitable for independent technical evaluation as an effective-model prototype. It should not yet be represented as an independently verified replacement for Lambda-CDM.